Does Weekly Aerobic Exercise Frequency Significantly Affect Daily Sleep Hours in High School Students?

An Empirical Analysis Based on YRBS 2007 Dataset (N = 12,042)

1. PROJECT OVERVIEW & MOTIVATION

Project Description

This project utilizes the 2007 YRBS dataset to analyze the relationship between **daily sleep duration** and **weekly aerobic exercise frequency** among US high school students.

Personal Inspiration

Inspired by personal experiences in heavy physical commitments like dance clubs or sports teams, this study investigates whether intense physical exhaustion compromises or promotes overall sleep health under the structural constraints of rigid academic schedules.

Dataset: Youth Risk Behavior Survey (YRBS) 2007 **Total Sample:** 12,042

2. HYPOTHESES & VARIABLES

Statistical Hypotheses

H_0: $\mu_{Low} = \mu_{Medium} = \mu_{High}$

(Exercise frequency has no impact on mean daily sleep hours.)

H_1: At least one group mean is different.

Variable Definitions

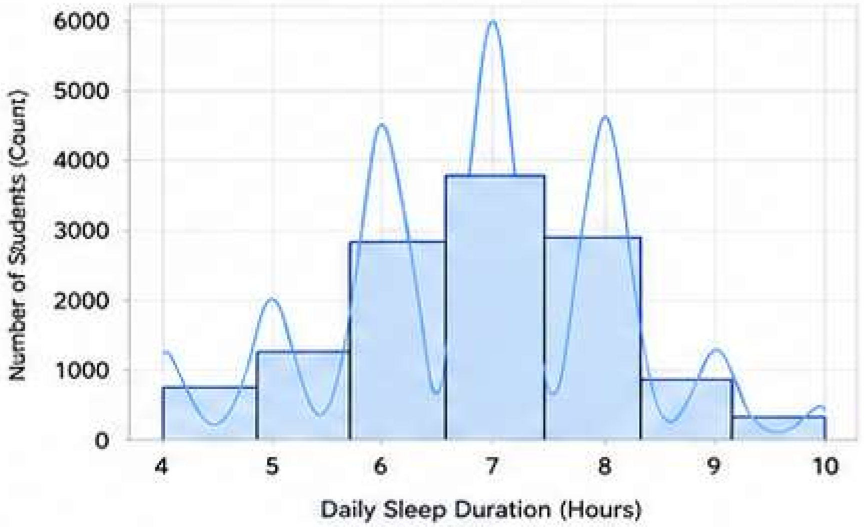
Predictor (X): **Exercise_Group** (days/week)

Low	Medium	High
1–2 days	3–4 days	≥ 5 days

Response (Y): **Sleep_Hours**

Actual daily sleep hours mapped into integer values from **4.0 to 10.0 hours.**

Distribution of Daily Sleep Hours (Normality Check)



Data satisfies the normality assumption for ANOVA due to the large sample size.

3. STATISTICAL RESULTS & DATA TABLES

Inferential Statistics (One-way ANOVA)

F-statistic: 41.2666

p-value: 1.3781×10^{-18} ($p < 0.05$)

→ Highly Significant! Reject H_0.

Tukey's HSD Post-hoc Comparisons

Comparison	Mean Diff (Hours)	p-value	Significant?
High vs. Low	+0.25	0.0000	Yes
High vs. Medium	+0.19	0.0000	Yes
Low vs. Medium	+0.06	0.1052	No

Descriptive Statistics (by Exercise Group)

	Low	Medium	High
Sample Size (N)	3201	3816	5025
Mean (Hours)	6.67	6.74	6.93
Std Dev	1.41	1.30	1.33
95% CI Lower Bound	6.670	6.736	6.925
95% CI Upper Bound	6.673	6.738	6.928

4. FINAL SUMMARY & DISCUSSION

Main Finding

Analyzing 12,042 samples robustly rejects the hypothesis that exercise compresses sleep.

- **High** (≥ 5 days/week): **6.93 hours** (highest)
- **Medium** (3–4 days/week): **6.74 hours**
- **Low** (1–2 days/week): **6.67 hours** (lowest)

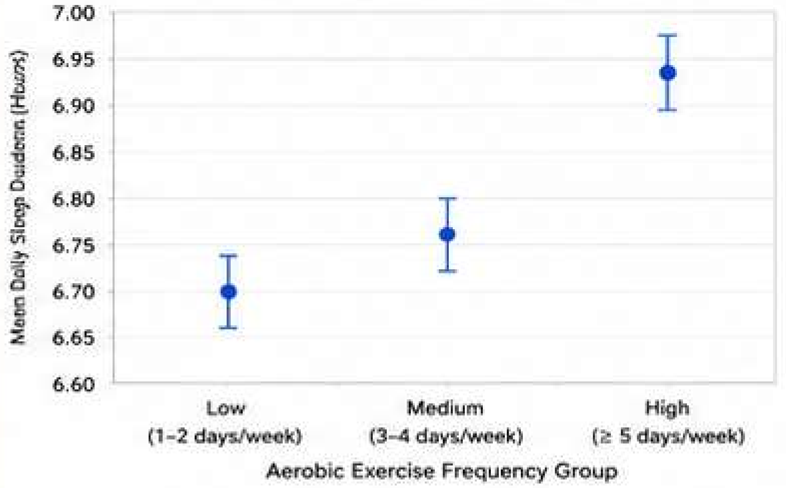
High-frequency exercisers slept the most.

Students exercising ≥ 5 days/week averaged 6.93 hours of sleep, compared with 6.67 hours among low-exercise peers.

Biological Interpretation

Frequent aerobic exercise may increase recovery-related sleep demand. Even under fixed school schedules, highly active students successfully negotiate and squeeze out approximately 15 to 20 extra minutes

95% CONFIDENCE INTERVAL PLOT



Intensive aerobic exercise is associated with more, not less, daily sleep among US high school students.